

Knapsack

wt $\rightarrow 1, 3, 4, 5$
val $\rightarrow 1, 4, 5, 7$
w $\rightarrow 7$



$3+4=7$
 $4+5=9$

optimize
 $\downarrow$
max profit

finite capacity

unbounded knapsack $\rightarrow$ infinite capacity

wt $\rightarrow 5, 5, 5, 7, 8, 9$
val $\rightarrow 12, 13, 15, 2, 3, 9$

wt $\rightarrow 22$

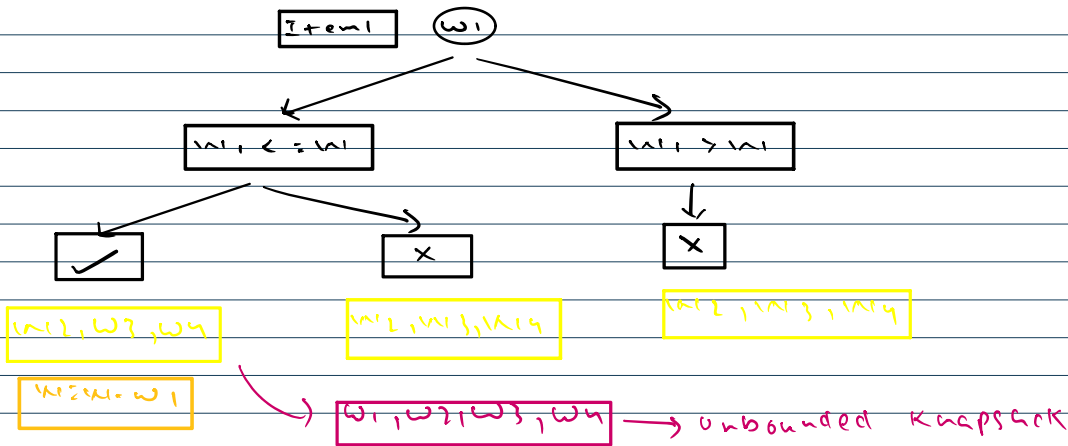
4 times $\rightarrow 5+5+5+5 = 20$

$15+15+15+15 = 60$

$5+5+5+7 \rightarrow 22+13+15+2 = 42$

Choice Diagram

Bag $\rightarrow w$



arr $\rightarrow 7, 1, 3, 6, 3, 4, 4, 8, 4$ $\rightarrow 34$

$17 \Rightarrow 2, 3, 4, 6, 4$
 $17 \Rightarrow 2, 3, 4, 8$

sum $\rightarrow$ even

Knapsack

if I'm able to fill the bag fully

if I'm able to fill the bag fully

choices
 $\checkmark$ $\times$
True $\rightarrow$ False $\rightarrow$ True

OR operator

$w = 0 \rightarrow$ return True
return False

True ~~False~~ $\Rightarrow$ True

True True $\Rightarrow$ True

False False $\Rightarrow$ False

$\vee$ OR
operator

between false

coin change problem

coins $\rightarrow 5, 2, 20, 10$

amount $\rightarrow 22$

How many ways you can provide him change?

$\left\{ \begin{array}{l} 5 + 5 + 5 + 5 + 2 \\ 5 + 5 + 10 + 2 \rightarrow \times \\ 10 + 10 + 2 \\ 20 + 2 \\ 2 \times 10 \\ 2 \times 6 + 10 \\ 2 \times 6 + 5 + 5 \\ 10 + 5 + 5 + 2 \rightarrow \times \end{array} \right. \quad (1) \Rightarrow \text{ways}$